Module 4. Special distribution

Discrete distributions	Continuous distributions
1. Binomial distribution	1.Uniform distribution (or) Rectangular distribution
2. Poisson distribution	2.Exponential distribution
3. Geometric distribution	3.Gamma distribution
4.Negative binomial distribution	4.Weibull distribution
	5.Normal distribution

Properties:

1. There must be a fixed number of trials.
2. All trials must have identical probabilities of success (P).
3. The trials must be independent of each other.

Notations:

n = number of trials
p = probability of success
q = probability of failure
X = A random variable which represents the number of successes.

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Applications:

Sampling process (or) Quality control measures in industries to classify the items as defective or non-defective.

Binomial distribution:

Probability mass function of Binomial distribution:

A discrete random variable X is said to follow binomial distribution if its probability mass function is

$$P(x) = n_{c_x} p^x q^{n-x}, \quad x = 0, 1, 2, \ldots\ldots. n$$

Moment generating function:

The moment generating function of a binomial variate is

$$M_x(t) = E[e^{tx}]$$

$$= \sum_{x=0}^{n} e^{tx} p(x)$$

$$= \sum_{x=0}^{n} e^{tx} n_{c_x} p^x q^{n-x}$$

$$= \sum_{x=0}^{n} n_{c_x} (pe^t)^x q^{n-x}$$

$$[(x+a)^n = n_{c_0} x^n + n_{c_1} x^{n-1} a + n_{c_2} x^{n-2} a^2 + \ldots\ldots\ldots + n_{c_n} a^n]$$

$$= (Pe^t + q)^n$$

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Mean and Variance of Binomial distribution:

$$\mu_1^1 = [\frac{d}{dt}\{M_x(t)\}]_{t=0}$$

$$= [\frac{d}{dt}\{(pe^t + q)^n\}]_{t=0}$$

$$= [n(pe^t + q)^{n-1} pe^t]_{t=0}$$

$$= n(p+q)^{n-1} p \qquad [\because p+q=1]$$

$$= np$$

Mean of the Binomial distribution is = np

$$\mu_2^1 = [\frac{d}{dt^2}^2 \{M_x(t)\}]_{t=0}$$

$$= [\frac{d^2}{dt^2}\{(pe^t + q)^n\}]_{t=0}$$

$$= [\frac{d}{dt}\{n(pe^t + q)^{n-1} pe^t\}]_{t=0}$$

$$= np[e^t(n-1)(pe^t + q)^{n-2} pe^t + (pe^t + q)^{n-1} e^t]_{t=0}$$

$$= np[(n-1)(p+q)^{n-2} p + (p+q)^{n-1}] \qquad [\because p+q=1]$$

$$= np[(n-1)p + 1]$$

$$\mu_2^1 = n(n-1)p^2 + np$$

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Variance of Binomial distribution:

$$\mu_2 = \mu_2^1 - (\mu_1^1)^2$$
$$= n(n-1)p^2 + np - n^2p^2$$
$$= n^2p^2 - np^2 + np - n^2p^2$$
$$= np(1-p)$$
$$= npq$$

Variance of Binomial distribution is npq

Problems:

1. **The mean of a binomial distribution is 5 and standard deviation is 2. Determine the distribution.**

Given mean = np = 5

S.D $= \sqrt{npq} =$ **2**

npq = 4

5q = 4

q = 4/5

p = 1-q=1/5

We have np = 5

n(1/5) = 5

n = 25

Hence the Binomial distribution is

$$p(X=x) = n_{C_x} p^x q^{n-x} = 25_{C_x} \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{25-x}, \quad x = 0,1,2......25.$$

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2. The mean and variance of a binomial distribution are 4 and 4/3. Find $P(X \geq 1)$

Given mean np = 4

Variance npq = 4/3

$$\frac{npq}{np} = \frac{4/3}{4} = \frac{1}{3}$$

q=1/3 ; p=2/3

np = 4

n(2/3) = 4

n = 6

$$p(X \geq 1) = 1 - P(X < 1)$$
$$= 1 - P(X = 0)$$
$$= 1 - 6_{C_0}\, p^0 q^{6-0}$$
$$= 1 - 6_{C_0}\,(2/3)^0 (1/3)^{6-0}$$
$$= 1 - (1/3)^6 = 0.998$$

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3. An irregular 6 faced die is such that the probability that it gives 3 even numbers in 5 throws is twice the probability that it gives 2 even numbers in 5 throws. How many sets of exactly 5 trials can be expected to give no even number out of 2500 sets?

Let X denote the number of even numbers obtained in 5 trias.

Given P(x=3)=2 *P(x=2)

$$5_{C_3}\, p^3 q^2 = 2 * 5_{C_2}\, p^2 q^3$$

$$p = 2q = 2(1-p) = 2 - 2p$$

$$3p = 2$$

i.e., $p = 2/3$

$$q = 1/3$$

Now, P(getting no even number)=P(x=0)

$$5_{C_0}\, p^0 q^5 = (1/3)^5 = \frac{1}{243}$$

Number of sets having no success (no even number) out of N sets=N *P(x=0)

Required number of sets=2500*1/243=10, nearly.

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